

Networking Assignment – 04

1. For the data bits 1011011:

- Add an even parity bit.
- Add an odd parity bit.
- If the receiver receives 10110111, determine whether an error is detected using even parity.
- Explain whether single parity checking can detect all types of errors.

2. A sender wants to transmit the 8-bit data:

11010110

- Determine the even parity bit and form the transmitted codeword.
- Determine the odd parity bit and form the transmitted codeword.
- If the receiver receives 11000110 instead of the original data, determine whether an error is detected using even parity.
- What happens if two bits are changed during transmission?

3. A sender wants to transmit the following four 8-bit words:

10101010 11001100 11110000 00001111

- Find the checksum.
- Show the complete data including the checksum.
- Explain how the receiver verifies whether an error occurred.

4. Consider the data: 1101011011 and the generator polynomial: 10011

Using CRC:

- Determine the degree of the generator polynomial.
- Append the required number of zeros to the data.
- Perform modulo-2 division.
- Find the CRC remainder.
- Write the final transmitted codeword.
- Explain how the receiver checks the received data.

5. CRC — Error Detection

A data bit stream 10011010 is transmitted using the standard CRC method. The generator polynomial is:
 $G(x) = x^3 + x + 1$

Determine the CRC remainder and the actual bit string transmitted. Show the complete modulo-2 division.

6. We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

7. Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

8. The signal-to-noise ratio is often given in decibels. Assume that SNR_{dB} = 36 and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

9. Shannon Capacity — SNR in dB

A communication channel has a bandwidth of 5 kHz and an SNR of 30 dB.

- Convert the SNR from dB

- b) Calculate the theoretical maximum channel capacity.
- c) Express your answer in kbps.

10. Network Delay

A packet of size 2 MB is transmitted over a link with a transmission rate of 20 Mbps. The distance between the sender and receiver is 1,500 km, and the propagation speed is:

$$2 \times 10^8 \text{ m/s}$$

The router also introduces:

- Processing delay = 2 ms
- Queuing delay = 8 ms

Calculate:

- a) Transmission delay
- b) Propagation delay
- c) Processing delay
- d) Queuing delay
- e) Total network delay

11. A bank sends a 1500-byte packet through a communication channel having:

$$\text{Bandwidth} = 10 \text{ kHz}$$

$$\text{SNR} = 20 \text{ dB}$$

$$\text{Number of signal levels} = 4$$

$$\text{Distance} = 1,000 \text{ km}$$

$$\text{Data rate} = 2 \text{ Mbps}$$

$$\text{Propagation speed} = 2 \times 10^8 \text{ m/s}$$

$$\text{Processing delay} = 1 \text{ ms}$$

$$\text{Queuing delay} = 4 \text{ ms}$$

Answer the following:

- a) Using Nyquist theorem, determine the maximum theoretical data rate.
- b) Convert 20 dB SNR into a linear SNR.
- c) Using Shannon theorem, determine the channel capacity.
- d) Determine the transmission delay of the 1500-byte packet.
- e) Determine the propagation delay.
- f) Determine the total network delay considering transmission, propagation, processing and queuing delays.
- g) Compare the Nyquist and Shannon results and explain why the practical maximum data rate cannot exceed the lower theoretical limit.